

Ethan Arnold NE 763 Exam 1

1. The individual layers of the TRISO particle are the kernel, the buffer, the IPyC, the SiC, and the OPyC. The kernel is the fuel and consists of Uranium (UO_2 , UCl_3 , $UClO$) that undergoes fission and produces heat. The buffer is a porous graphite layer that is designed to mechanically decouple the kernel and PyC layers and accommodate swelling and fission gas. The inner Pyrolytic carbon layer (IPyC) protects the SiC from fission product attack and puts the SiC in compression. The SiC is a layer of Silicon Carbide that is the main fission product barrier and pressure containment. The OPyC is the outer Pyrolytic carbon layer that also puts the SiC in compression and protects it during manufacturing. 10/10
2. Carbides have excellent high temperature compatibility with graphite. However, they have a tendency for rare earth fission products to migrate through the buffer and attack the SiC. Oxides on the other hand, do not see as much fission product corrosion of the outer carbide layers, but they suffer from increased carbon monoxide (CO) production due to higher oxygen potential (partial oxygen pressure). This greatly increases pressure in the particle and even dominates the pressure contribution at higher burnups. Oxides do not oxidize during manufacturing and are stable up to higher temperatures. VCO offers a compromise by retaining fission products but creating a solid oxide as a result of oxidation with UCl_2 . 12/12
3. Fission product chemistry changes based on fission product solubility. Soluble fission products may oxidize and stay within the kernel, however, insoluble fission products may precipitate to grain boundaries and be released from the kernel. Palladium is an example of an insoluble fission product that corrodes carbide layers. Oxides are generally better at retaining fission products. As mentioned, oxygen can oxidize carbon and result in CO release. With VCO, the free oxygen oxidizes the UCl_2 first resulting in a solid oxide and then oxidizes fission products which are otherwise carbides. Fission products are either insoluble oxides, soluble oxides, metals, volatiles, or noble gases. 8/8

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14. Irradiation damage accumulation causes swelling of the SiC. In low temperature regimes, damage largely results in dark spot ~~defect~~ interstitial clusters. With enough accumulation, these clusters can cause swelling that forces the SiC to be amorphous. Above the critical amorphization temperature, point defect swelling dominates. The swelling from these point defects is logarithmic until it reaches saturation. The swelling saturation point is lower with increasing temperature. At high temperature, Frank interstitial loops form that refuse to glide (sessile). Vacancies and voids continue to form and now void swelling dominates. Void swelling does not saturate like point defect swelling but continues to increase with dose. The irradiation damage in the low T regime is also logarithmic with dose until amorphization. The low T regime can result in very significant swelling up to 10%.

Thermal conductivity in SiC is generally high. However, with irradiation, the thermal conductivity decreases. Thermal conductivity degradation is dominated by defects at low temperatures and saturation is reached eliminating temperature dependence. When void swelling occurs, thermal conductivity is not as impacted as defects are the main phonon scatterers. Voids do not scatter phonons as effectively. Thermal conductivity continues to decrease with dose and increase with temperature but the relationship is not linear. Thermal defect resistance is linear with swelling and allows for estimates based on SiC density.

- wanted some more

Graphite behaves uniquely as a polycrystalline, anisotropic material. The coefficient of thermal expansion is much higher in the c plane than the basal plane (a) and when heat treated and cooled, Mizoroki cracks form in the ~~in~~ the basal plane. Then when defects form between planes, the expansion in the c direction results in a contraction in a to fill in cracks and vacancies. Thus, graphite shrinks initially under irradiation. Eventually, the cracks and vacancies are filled and a turnaround occurs where graphite begins to expand with increased irradiation.

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7. 1) Overpressure - overpressure occurs when fission product buildup and CO buildup create too much tensile stress in the SiC and it bursts
- 2) Kernel coating Material Interaction (KAMI) - occurs when the kernel comes in contact with a dense coating layer. The SiC typically fails quickly after this. The kernel contact occurs as the kernel swells.
- 3) Kernel migration (ambipolar effect) - occurs when the kernel shifts away from the center and towards a boundary layer. It occurs due to a thermal gradient which carbon preferentially diffuses up while the kernel - diffuses down the gradient. It is essentially a product of carbon transport.

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9. Zirconium carbide is an advanced TRISO concept of interest. It may exhibit excellent thermal conductivity and stability. It does not allow Palladium to escape (although it does have some issues with Rhenium escape). It is thermally stable up to higher temperatures than SiC (2400°C whereas SiC breaks down at 2000°C). It does not solve the Ar escape problem. It is less oxidation resistant than SiC but does not show the same thermal degradation. It may also be added as a layer around the kernel to retain fission products and reduce swelling.

5/5

9. Corrosion and creep are the major concerns for I AX materials in a He environment. Nickel alloys are being looked at for their high temperature performance and ability to have passive chromia layers to protect against corrosion. Corrosion and creep are issues due to long lifetimes at high temperatures. Corrosion is broken into three regimes: A chemical regime where oxidation occurs at all surfaces throughout the material (open surfaces) at low temperature, an in pore regime where oxidation occurs faster but is not transported as far in the material, and a surface layer regime where oxidation occurs rapidly at the surface at high temperature.
- this is oxidation of graphite

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9 cont'd: The surface layer reaction rate is too high for oxygen to diffuse through the material. Corrosion largely occurs due to impurities in the helium which is why highly pure He is needed (< 5 ppm impurities). Molecular sieves help remove these impurities. Also, the alloys being examined include carbides in their grains to reduce creep. The ideal alloys are materials with the ability to form passive, self-healing oxide layers.

- Still covered a lot, but included graphite behavior...